

Vineth Joseph Choudary Mallavarapu

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SUMMARY

M.Sc. student in Mechatronics, Robotics and Biomechanical Engineering at TUM with hands-on experience in ROS 2, SLAM, embedded systems, UAV operations, and industrial robotics. Seeking a working student role in robotics, automation, mechatronics, or embedded systems.

EDUCATION

Technical University of Munich	Oct 2025 – Present
M.Sc. in Mechatronics, Robotics and Biomechanical Engineering	
SRM Institute of Science and Technology	2020 – 2024
B.Tech. in Mechatronics Engineering with specialization in Robotics	CGPA: 8.16/10

INDUSTRIAL EXPERIENCE

Grace Drone Technologies — Drone Operations Support Engineer	Oct 2024 – Sep 2025
- Supported UAV mission planning, aerial inspections, and mapping operations using DJI Matrice 30T and DroneDeploy. - Processed aerial imagery to generate orthomosaics, site layouts, and survey outputs using AutoCAD, and assisted with field operations, safety checks, and data handling.	
Fronius India Pvt. Ltd. — Robotics and Welding Automation Intern	May 2024 – Jun 2024
- Conducted experiments on a FANUC 6-axis industrial robot to evaluate welding modes including CMT, PMC, Pulse, and LSC. - Analyzed process parameters and weld quality to support manufacturing and process optimization.	

PROJECTS

Autonomous Mobile Robot using ROS 2 and SLAM	2025
- Developed an autonomous mobile robot using ROS 2 Jazzy, implementing SLAM for real-time mapping and navigation. - Integrated a LLaMA-based LLM for voice and chat-based interaction, enabling natural language command execution. - Developed and tested the system on a differential-drive platform using Python and ROS 2.	
Automated Millet Identification and Billing System	2024
- Built an automated millet classification system that uses deep learning to identify millet varieties through an overhead camera with 85% accuracy, coupled with load cells for weight estimation. - The system autonomously generates a bill based on the measured data. - The system also includes a touch display with a Tkinter-based user interface and an admin portal for updating prices and viewing billing history.	
Robotic Conveyor Sorting and Assembly Workcell	2023
- Fabricated and developed a robotic workcell for colour-based object sorting and assembly, featuring an RRR manipulator with inverse kinematics and a conveyor system. - When the system senses a black cube, the manipulator is activated to pick the object and place it in one bin. For a white cube, an actuator mounted above the conveyor pushes the object into another bin. - Integrated an IoT-based control system using ESP32.	

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

Robotics & Automation: ROS 2, SLAM, Robot Kinematics and Dynamics, Industrial Manipulators, Robotic Welding, IoT Systems

Embedded Platforms: Arduino, Raspberry Pi, ESP32

CAD & Simulation: SolidWorks, ANSYS, AutoCAD

Tools: Git, Linux, DroneDeploy

PATENT / PUBLICATION

Patent: System for Identifying and Weighing Loose Items (Millets), App. No. 202541006107 (Published Jan 2025)

Publication: "Development of Automated Millet Identification and Billing System," ADCIS 2024 (Springer)

LANGUAGE / LEADERSHIP

Languages: English, German (A1)

Powertrain Engineer — Infieon Super Mileage (Shell Eco-Marathon Team): Contributed to energy-efficient propulsion system design within a multidisciplinary team

Association of Mechatronics — Social Service Lead: Coordinated student events and community initiatives